

Module 4 Lesson 4

Compound Interest (8.4)

Objectives

In this lesson, we will learn how to

- use compound interest formulas;
- calculate present value;
- understand and compute effective annual yield.

1. Compound Interest

Compound interest is interest computed on the original principal as well as on any accumulated interest.

This means that interest earns more interest over time.

Simple Interest vs. Compound Interest

- Simple interest is calculated only on the original principal.
- Compound interest is calculated on the principal and on previously earned interest.

2. Compound Interest Paid Once a Year

If interest is compounded once per year, the future value is given by

$$A = P(1 + r)^t.$$

Compound Interest Formula: Once Per Year

$$A = P(1 + r)^t$$

where

A = future value,

P = principal or present value,

r = annual interest rate as a decimal,

and

t = time in years.

3. Using the Compound Interest Formula

Example 1: You deposit \$2000 in a savings account at Hometown Bank. The account has an annual interest rate of 1.6%.

- (a) Find the amount, A , of money in the account after 3 years, subject to compound interest.
- (b) Find the interest.

Solution

Here,

$$P = 2000, \quad r = 1.6\% = 0.016, \quad t = 3.$$

Use

$$A = P(1 + r)^t.$$

$$A = 2000(1 + 0.016)^3$$

$$A = 2000(1.016)^3$$

$$A \approx 2097.54.$$

Rounded to the nearest cent, the amount in the savings account after 3 years is

$$\$2097.54.$$

The interest is the future value minus the principal:

$$2097.54 - 2000 = 97.54.$$

Thus, the interest earned after 3 years is

$$\$97.54.$$

4. Compound Interest Paid n Times Per Year

Sometimes interest is compounded more than once per year.

For example:

- monthly compounding means $n = 12$;
- quarterly compounding means $n = 4$;

- daily compounding means $n = 365$.

Compound Interest Formula

If P dollars are deposited at annual interest rate r , compounded n times per year, then the amount A after t years is

$$A = P \left(1 + \frac{r}{n} \right)^{nt}.$$

Here, A is the account's future value, and P is its present value.

5. Monthly Compounding

Example 2: You deposit \$7500 in a savings account that has an annual interest rate of 1.2%. The interest is compounded monthly.

- (a) How much money will you have after 5 years?
- (b) Find the interest after 5 years.

Solution

Here,

$$P = 7500, \quad r = 1.2\% = 0.012, \quad t = 5, \quad n = 12.$$

Use

$$A = P \left(1 + \frac{r}{n} \right)^{nt}.$$

$$A = 7500 \left(1 + \frac{0.012}{12} \right)^{12(5)}$$

$$A = 7500(1.001)^{60}$$

$$A \approx 7963.54.$$

Rounded to the nearest cent, you will have

$$\$7963.54$$

after 5 years.

The interest is

$$7963.54 - 7500 = 463.54.$$

Thus, the interest earned after 5 years is

$$\$463.54.$$

6. Continuous Compounding

Some banks use continuous compounding, where the number of compounding periods increases indefinitely.

Continuous Compounding Formula

For continuous compounding, the future value is

$$A = Pe^{rt},$$

where

$$e \approx 2.71828.$$

Two Compound Interest Formulas

For n compounding periods per year:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}.$$

For continuous compounding:

$$A = Pe^{rt}.$$

7. Choosing Between Investments

Example 3: You decide to invest \$8000 for 6 years and have a choice between two accounts.

- Account 1 pays 2.5% per year, compounded monthly.
- Account 2 pays 2.49% per year, compounded continuously.

Which is the better investment?

Solution

The better investment is the one with the greater balance after 6 years.
For Account 1:

$$P = 8000, \quad r = 0.025, \quad n = 12, \quad t = 6.$$

$$A = 8000 \left(1 + \frac{0.025}{12} \right)^{12(6)}$$

$$A \approx 9293.22.$$

The balance in Account 1 after 6 years is

$$\$9293.22.$$

For Account 2:

$$P = 8000, \quad r = 0.0249, \quad t = 6.$$

$$A = Pe^{rt}$$

$$A = 8000e^{0.0249(6)}$$

$$A \approx 9289.10.$$

The balance in Account 2 after 6 years is

$$\$9289.10.$$

Since

$$9293.22 > 9289.10,$$

the better investment is the 2.5% account compounded monthly.

8. Present Value

Sometimes we know how much money we want in the future, and we need to find how much to invest now.

This amount is called the present value.

Present Value Formula

If A dollars are to be accumulated in t years in an account that pays annual rate r , compounded n times per year, then the present value P is

$$P = \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}}.$$

9. Calculating Present Value

Example 4: How much money should be deposited today in an account that earns 6% compounded monthly so that it will accumulate to \$20,000 in 5 years?

Solution

Here,

$$A = 20,000, \quad r = 6\% = 0.06, \quad n = 12, \quad t = 5.$$

Use the present value formula:

$$\begin{aligned} P &= \frac{A}{\left(1 + \frac{r}{n}\right)^{nt}} \\ P &= \frac{20,000}{\left(1 + \frac{0.06}{12}\right)^{12(5)}} \\ P &= \frac{20,000}{(1.005)^{60}} \\ P &\approx 14,827.45. \end{aligned}$$

Approximately

$$\$14,827.45$$

should be invested today to accumulate to \$20,000 in 5 years.

10. Effective Annual Yield

The **effective annual yield**, or **effective rate**, is the simple interest rate that produces the same amount of money at the end of one year as a compound interest account.

Effective Annual Yield

For an annual interest rate r compounded n times per year, the effective annual yield is

$$\left(1 + \frac{r}{n}\right)^n - 1.$$

11. Understanding Effective Annual Yield

Example 5: You deposit \$4000 in an account that pays 4% interest compounded monthly.

- (a) Find the future value after 1 year.
- (b) Use the future value formula for simple interest to determine the effective annual yield.

Solution

Here,

$$P = 4000, \quad r = 4\% = 0.04, \quad n = 12, \quad t = 1.$$

Using compound interest,

$$A = 4000 \left(1 + \frac{0.04}{12} \right)^{12(1)}.$$

$$A = 4000 (1.003333\dots)^{12}$$

$$A \approx 4162.97.$$

So the account's future value after one year is

$$\$4162.97.$$

The interest earned is

$$4162.97 - 4000 = 162.97.$$

Now use the simple interest formula

$$I = Prt.$$

Since $t = 1$,

$$162.97 = 4000(r)(1).$$

$$r = \frac{162.97}{4000}$$

$$r \approx 0.0407.$$

Convert to a percent:

$$0.0407 = 4.07\%.$$

Thus, the effective annual yield is approximately

$$4.07\%.$$

This means that 4% compounded monthly is equivalent to about 4.07% simple interest for one year.

Summary

Key Ideas

- Compound interest is calculated on both the principal and previously earned interest.

- If interest is compounded once per year, use

$$A = P(1 + r)^t.$$

- If interest is compounded n times per year, use

$$A = P \left(1 + \frac{r}{n} \right)^{nt}.$$

- For continuous compounding, use

$$A = Pe^{rt}.$$

- Present value tells how much must be invested now to reach a future amount.

- The present value formula is

$$P = \frac{A}{\left(1 + \frac{r}{n} \right)^{nt}}.$$

- Effective annual yield gives the equivalent simple interest rate for one year.